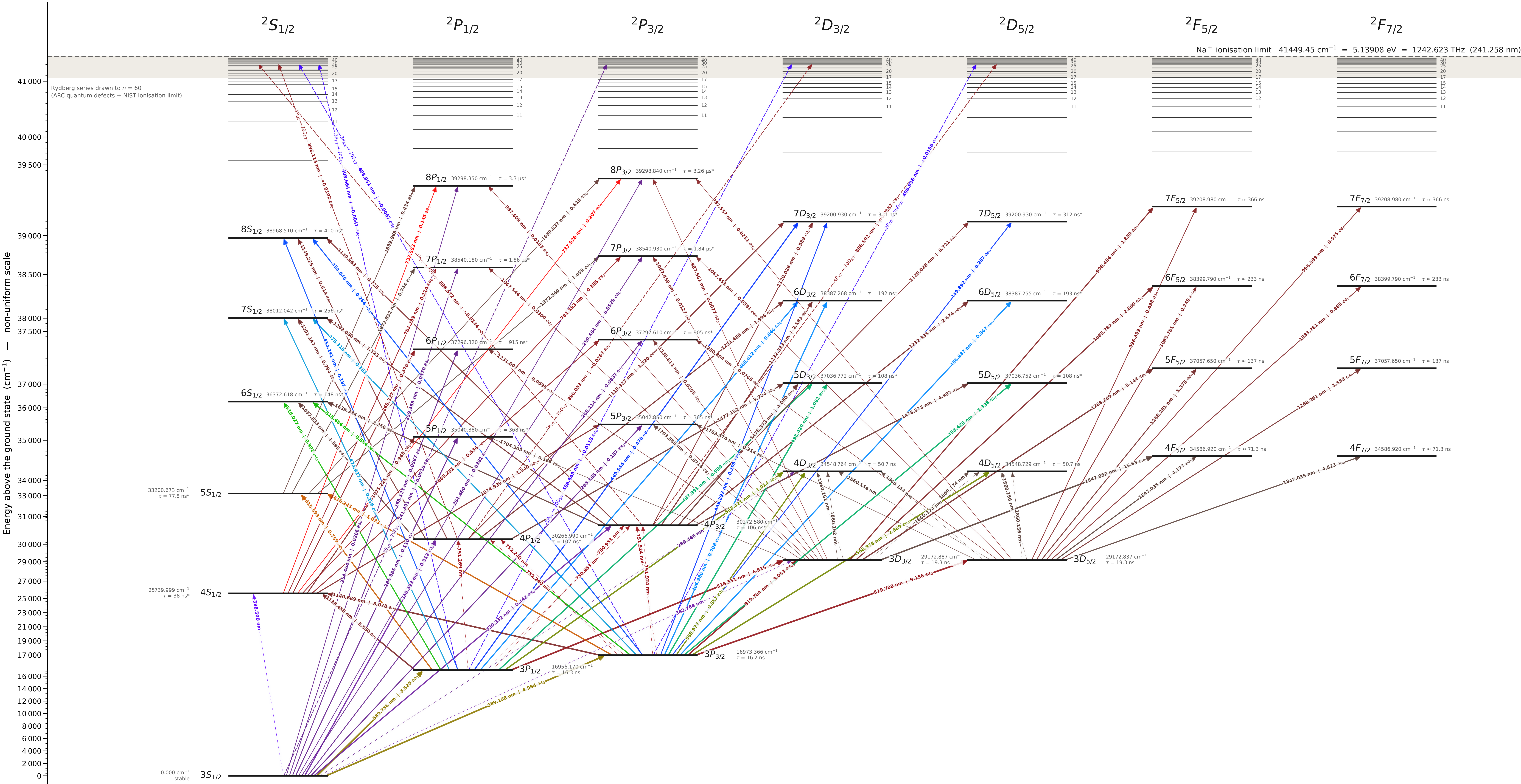
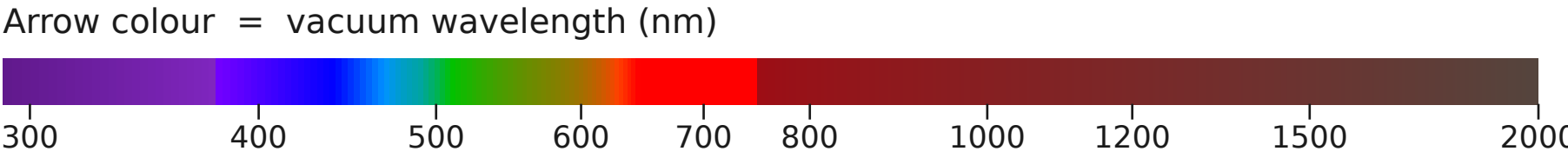


Sodium-23

²³Na (Na I), nuclear spin $I = 3/2$. Energy levels and transitions below 2 μm : 36 levels, 106 lines.

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle J||er||J' \rangle$ in $e a_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and lifetime (measured where a value exists, otherwise ARC, marked $\approx$).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\tilde{\nu}$ (cm^{-1})	$\langle J er J' \rangle$ ($e a_0$)	A (s^{-1})	$\Gamma/2\pi$ up. (MHz)	branch (%)	Typical use
$3S_{1/2} \rightarrow 3P_{3/2}$	589.1583	588.9951	508.8487	16973.366	4.984	6.152e+07	9.795	100.0	D2 line
$3S_{1/2} \rightarrow 3P_{1/2}$	589.7558	589.5924	508.3332	16956.170	3.525	6.135e+07	9.765	100.0	D1 line
$3P_{3/2} \rightarrow 3D_{5/2}$	819.7077	819.4824	365.7309	12199.471	9.156	5.140e+07	8.259	100.0	
$3P_{1/2} \rightarrow 3D_{3/2}$	818.5505	818.3255	366.2480	12216.717	6.815	4.290e+07	8.259	83.4	
$3P_{3/2} \rightarrow 4S_{1/2}$	1140.6888	1140.3766	262.8170	8766.633	5.078	1.760e+07	4.187	66.6	
$3P_{3/2} \rightarrow 4D_{5/2}$	568.9783	568.8205	526.8961	17575.363	2.569	1.210e+07	3.139	63.5	
$3P_{1/2} \rightarrow 4D_{3/2}$	568.4210	568.2633	527.4127	17592.594	1.914	1.010e+07	3.139	53.0	
$3P_{1/2} \rightarrow 4S_{1/2}$	1138.4557	1138.1441	263.3326	8783.829	3.580	8.800e+06	4.187	33.4	
$3P_{3/2} \rightarrow 3D_{3/2}$	819.7043	819.4790	365.7324	12199.521	3.053	8.570e+06	8.259	16.6	
$3P_{3/2} \rightarrow 5S_{1/2}$	616.2452	616.0747	486.4824	16227.307	1.073	4.980e+06	2.045	38.7	
$3P_{3/2} \rightarrow 5D_{3/2}$	498.4199	498.2808	601.4858	20063.406	1.092	4.880e+06	1.472	8.8	
$3P_{3/2} \rightarrow 5D_{5/2}$	498.4204	498.2813	601.4852	20063.386	1.338	4.880e+06	1.468	52.9	
$3P_{1/2} \rightarrow 5D_{3/2}$	497.9930	497.8541	602.0013	20080.602	0.999	4.090e+06	1.472	44.2	
$3S_{1/2} \rightarrow 4P_{3/2}$	330.3319	330.2369	907.5491	30272.580	0.442	2.750e+06	1.501	28.2	330 nm UV line: narrow-line cooling / UV MOT
$3S_{1/2} \rightarrow 4P_{1/2}$	330.3929	330.2978	907.3815	30266.990	0.312	2.730e+06	1.490	28.0	330 nm UV line
$3P_{1/2} \rightarrow 5S_{1/2}$	615.5929	615.4225	486.9979	16244.503	0.759	2.500e+06	2.045	19.4	
$3P_{3/2} \rightarrow 6D_{3/2}$	466.9864	466.8557	641.9726	21413.902	0.708	2.490e+06	0.827	8.0	

λ and ν from NIST level energies (centre of gravity, no hyperfine structure); λ air uses standard air (Ciddor).
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$. Branch: share of the upper level's decays that takes this line.

Rydberg excitation wavelengths (vacuum, nm)

Lower $\rightarrow$ series	n = 30	n = 40	n = 50	n = 70	n = 100	n $\rightarrow$ ∞	d% ARC
$3S_{1/2} \rightarrow nP_{3/2}$	242.012	241.675	241.522	241.391	241.323	241.258	0.0011
$3P_{3/2} \rightarrow nS_{1/2}$	410.806	409.792	409.337	408.951	408.750	408.562	0.0067
$3P_{3/2} \rightarrow nD_{5/2}$	410.610	409.711	409.297	408.936	408.745	408.562	0.0158
$4S_{1/2} \rightarrow nP_{3/2}$	641.837	639.474	638.406	637.491	637.012	636.559	0.0044
$3D_{5/2} \rightarrow nP_{3/2}$	823.219	819.336	817.583	816.083	815.298	814.557	0.0003
$3D_{5/2} \rightarrow nF_{7/2}$	822.729	819.133	817.480	816.046	815.286	814.557	0.0457
$4P_{3/2} \rightarrow nS_{1/2}$	905.535	900.623	898.431	896.572	895.608	894.705	0.0144
$4P_{3/2} \rightarrow nD_{5/2}$	904.583	900.233	898.235	896.502	895.584	894.705	0.0357
$5S_{1/2} \rightarrow nP_{3/2}$	1231.589	1222.918	1219.015	1215.683	1213.944	1212.301	0.0096
$4D_{5/2} \rightarrow nP_{3/2}$	1476.769	1464.320	1458.728	1453.960	1451.472	1449.124	0.0023
$4D_{5/2} \rightarrow nF_{7/2}$	1475.191	1463.672	1458.401	1453.842	1451.432	1449.124	0.0847
$4F_{7/2} \rightarrow nD_{5/2}$	1483.574	1471.910	1466.575	1461.961	1459.523	1457.188	0.0087
$5P_{3/2} \rightarrow nS_{1/2}$	1594.151	1578.993	1572.267	1566.583	1563.642	1560.890	0.0244
$5P_{3/2} \rightarrow nD_{5/2}$	1591.204	1577.794	1571.665	1566.368	1563.569	1560.890	0.0612
$6S_{1/2} \rightarrow nP_{3/2}$	2021.162	1997.914	1987.519	1978.677	1974.073	1969.732	0.0167

Rydberg term energies from ARC quantum defects + NIST ionisation limit; lower level from NIST. $d_{70} = \langle J||er||J' \rangle$ to $n = 70$ in $e a_0$ (calculated).
 $S \rightarrow nP$ values come from a near-cancelling integral: treat them as an order of magnitude.

²³Na hyperfine structure ($I = 3/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$3S_{1/2}$	885.813064		- F=1: -1107.27 F=2: +664.36
$3P_{1/2}$	94.390(20)		- F=1: -117.99 F=2: +70.79
$3P_{3/2}$	18.532(6)	2.722(7)	F=0: -66.09 F=1: -50.28 F=2: -15.94 F=3: +42.38
$4S_{1/2}$	203.60(20)		- F=1: -254.50 F=2: +152.70
$3D_{5/2}$	0.1085(24)		- F=1: -0.57 F=2: -0.35 F=3: -0.03 F=4: +0.41
$3D_{3/2}$	0.527(25)		- F=0: -1.98 F=1: -1.45 F=2: -0.40 F=3: +1.19
$4P_{1/2}$	30.60(10)		- F=1: -38.25 F=2: +22.95
$4P_{3/2}$	6.02(6)	0.97(6)	F=0: -21.37 F=1: -16.32 F=2: -5.24 F=3: +13.79
$5S_{1/2}$	77.40(14)		- F=1: -96.75 F=2: +58.05
$4D_{5/2}$	0.029(6)		- F=1: -0.15 F=2: -0.09 F=3: -0.01 F=4: +0.11
$4D_{3/2}$	0.215(15)		- F=0: -0.81 F=1: -0.59 F=2: -0.16 F=3: +0.48
$5P_{1/2}$	13.469(4)		- F=1: -16.84 F=2: +10.10
$5P_{3/2}$	2.6360(23)	0.370(8)	F=0: -9.42 F=1: -7.16 F=2: -2.25 F=3: +6.02

A, B: magnetic-dipole and electric-quadrupole constants (measured; see data file for sources).